



PROJECT 4 · 2102571 MULTIMEDIA COMMUNICATION

# Spatial Audio & High-Fidelity Streaming

*Evaluating Opus, AAC and MP3 for low-latency binaural audio over constrained networks*

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# Why Spatial Audio?

*The bandwidth gap that quietly breaks immersion*



## Bandwidth gap

Video uses 3-8 Mbps;  
audio is capped at 64-128 kbps.



## Spatial cues are fragile

ITD and ILD live in transients and  
high frequencies — the first to be cut.



## Immersion at risk

VR / AR / Metaverse demand  
presence — bad audio breaks it.

### Question

**Which codec converts the smallest bitrate into the most directional clarity?**

# Objectives

*Four concrete questions, one experimental program*

01

## Compare codecs

MP3 vs AAC vs Opus across 24-256 kbps

02

## Find transparency

Bitrate at which listeners can no longer tell

03

## Build the player

Web Audio API + HRTF, deployed and live

04

## Stress the network

Loss / jitter → spatial breakdown point

# System & Methodology

*End-to-end: encode → analyse → spatialise → stress*



## FFmpeg

libfdk-aac · libmp3lame ·  
libopus



## Metrics

SNR + PESQ-derived ODG



## Spatial player

Web Audio API + HRTF



## Stress sim

Packet loss + jitter sliders

## ABX listening test

Forced choice: A vs B vs X.  
3 codecs × 5 bitrates × 5 reps = 75 trials.  
Web interface; CSV log per participant.

## Stress design

Loss: 0–20 % · Jitter: 0–200 ms.  
Mono / Stereo / HRTF compared.  
Metric = Immersion Degradation Index.

# Codec Comparison Results

*Rate-distortion: SNR misleads, ODG tells the truth*

[ Figure — Rate-Distortion (ODG) ]  
drop ./report/figures/rd\_curve\_odg.png

*Figure 5 — ODG vs. bitrate (PESQ-derived)*

**48 kbps**

Opus transparency (pilot)

**64 kbps**

MP3 / AAC transparency

**≈ 20 ms**

Opus algorithmic delay

# ABX Test & Transparency

*Pilot data — full cohort study in progress*

| Codec | Transparency bitrate | ABX accuracy at threshold | MOS at threshold |
|-------|----------------------|---------------------------|------------------|
| MP3   | ≈ 64 kbps            | 54 % (≈ chance)           | MOS 4.3          |
| AAC   | ≈ 64 kbps            | 53 % (≈ chance)           | MOS 4.4          |
| Opus  | ≈ 48 kbps            | 52 % (≈ chance)           | MOS 4.5          |

## Takeaway

Opus reaches transparency roughly 1.3× faster than MP3/AAC.  
All three converge on near-chance discrimination at 64 kbps and above.

# Network Stress: Spatial Breaks First

*HRTF binaural streams are roughly 2× more fragile than stereo*

**> 20 %**

Mono — packet loss tolerated

**≈ 15 %**

Stereo — packet loss tolerated

**≈ 10 %**

Spatial (HRTF) — breakdown

## Why?

HRTF rendering relies on phase-coherent two-channel delivery.  
A single late or dropped packet disrupts ITD/ILD and momentarily collapses the scene.

LIVE DEMO

# Spatial Audio Lab

[vanowarna.com/spatial-audio-and-high-fidelity-streaming](https://vanowarna.com/spatial-audio-and-high-fidelity-streaming)



## Drop a sound source

Drag the orb in 3D space — listener stays centred



## Switch codecs live

Original ↔ MP3 ↔ AAC ↔ Opus mid-playback



## Inject network stress

Loss & jitter sliders — feel the spatial collapse

# Conclusions

*Pick Opus, ship spatial, design for fragility*

## What we found



### Opus wins

Best quality / latency / bitrate balance



### Spatial fragility

HRTF breaks at ~10 % loss; stereo holds longer



### Bandwidth myth

5G has the bytes — jitter & loss are the real risk

## Where this goes next



MPEG-H 3D Object-Based Audio



Personalised HRTFs (CNN-based)



Opus multistream for ambisonics



Neural codecs (EnCodec / SoundStream)



WebRTC pipeline on real 4G/5G traces

# Thank you

*Questions?*



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